

13-12-2018

Rehearsal Examination 2018 Marking Scheme for class - XII Subject - Physics

Section A:

- 1a) (i) op (A) (ii) op (C) $\rightarrow 3.4 \text{ eV}$ (iii) $eV_0 = h\nu - h\nu_0$ thus,
($470 \pm 5\% \text{ nm}$) as $|E_T| = E_K$ or $V_0 = \frac{h}{e} \nu - \frac{h}{e} \nu_0$
slope $= \frac{h}{e}$; $C = \frac{-h\nu_0}{e}$
 $\therefore \left| \frac{C}{\text{slope}} \right| = \nu_0$ op (D)
- (iv) op (B) opposite phase
(v) op (B) : $\neq 0$ volt.

1b) i) For Ryman Series : $\frac{1}{\lambda} = R_H \left(\frac{1}{1^2} - \frac{1}{n_i^2} \right)$

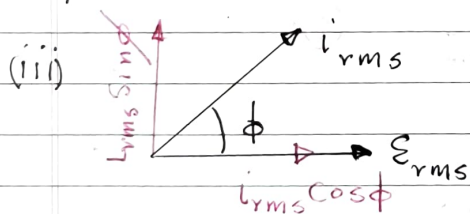
For shortest λ , $n_i = \infty$

or $\frac{1}{\lambda} = 1.097 \times 10^7 \left[1 - \frac{1}{\infty^2} \right]$

$\therefore \lambda = 0.912 \times 10^{-7} \text{ m} = 912 \text{ \AA}$

(vii) $\oint_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 \left[I_c + \epsilon_0 \frac{d\phi_E}{dt} \right]$
 $I_c \rightarrow$ conduction current
 $\phi_E \rightarrow$ Electric flux.

- (ii) This happens when the emf of the driving auxillary battery becomes less than the emf of the cell to be measured.



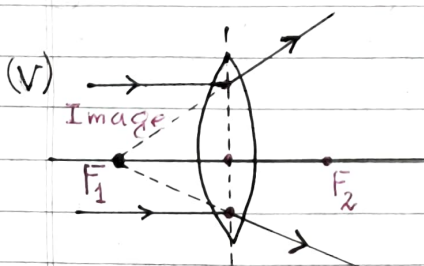
$\therefore$ Power contributed by

$i_{rms} \sin \phi$ is:

$\epsilon_{rms} (i_{rms} \sin \phi) \cos(90^\circ)$

$\therefore i_{rms} \sin \phi$ is Watt-less current.

(iv) $Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{25} \sqrt{\frac{10}{40 \times 10^{-6}}} = \frac{1000}{50} = 20$



vi) Primary Rainbow : Violet inner edge
red colour on the outer edge.

Secondary Rainbow : Red inner edge
violet colour on the outer edge

Section B

2) $j = ne v_d$; $\vec{j} = \sigma \vec{E}$

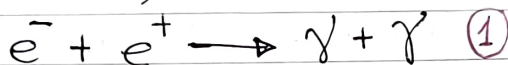
OR

2) $\phi = 5t^3 + t^2 - 5$

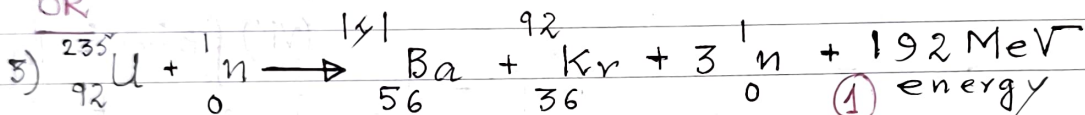
$\epsilon = \frac{-d\phi}{dt} = -15t^2 - 8t$ at $t = 2s$

$\epsilon = -15 \times 2^2 - 8 \times 2 = -76 \text{ volt}^-$

3) ~~When an electron (e^-) and a positron (e^+) collide at low energy pair annihilation occurs followed by the creation of γ -ray photons.~~



OR



Control rods $\rightarrow$ absorb some of the neutrons produced during chain reaction thereby resulting in controlled fission.

4) By ECP:

$$\frac{1}{2} K (r)^2 = \frac{q^2}{4\pi\epsilon} \left[\frac{1}{r} - \frac{1}{2r} \right]$$

or $\frac{K r^2}{2} = \frac{q^2}{4\pi\epsilon} \left(\frac{2-1}{2r} \right)$

or $K = \frac{q^2}{4\pi\epsilon} \times \frac{1}{r^3}$

5) $M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{2} \right) = -\frac{150}{5} \left(1 + \frac{5}{25} \right) = -33.6$

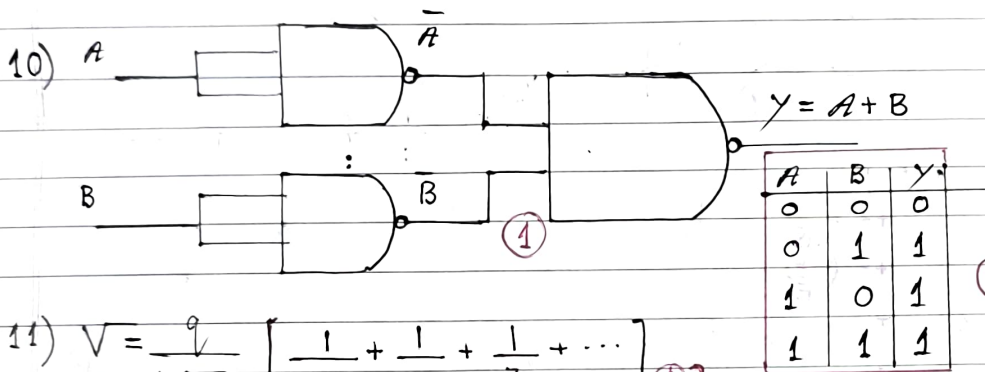
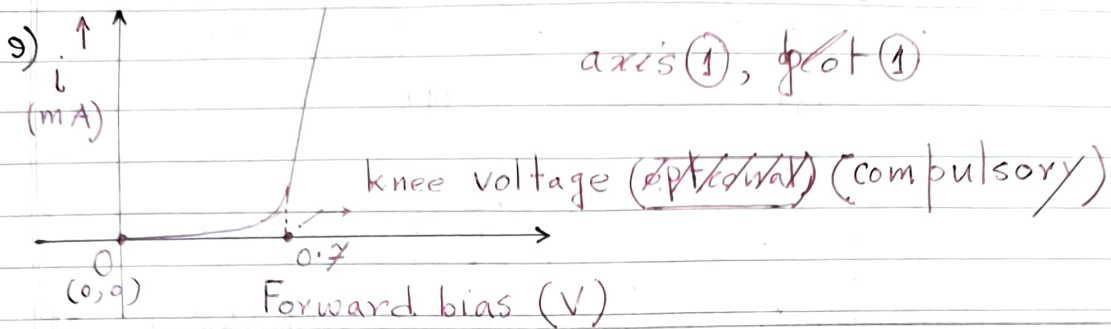
6) 'Interference' is the result of interaction of light waves coming from two different wavefronts originating from 2 coherent sources.

Diffraction is the result of interaction of light waves coming from 2 parts of same wave front.

(1+1)

7) $\beta = \frac{D\lambda}{d}$, $\beta' = \frac{D\lambda}{d\mu}$ (1) or $\beta' = \frac{\beta}{\mu} = \frac{2 \text{ mm}}{4/3} = 1.5 \text{ mm}$ (1)

8) $\beta = \left(\frac{\alpha}{1-\alpha} \right) \Rightarrow \alpha = \frac{\beta}{1+\beta}$ or $\alpha = 0.986$



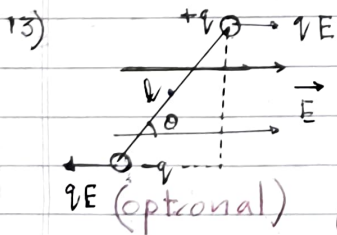
$$11) \quad V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots \right] \quad (1)$$

$$E = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right] \quad (1)$$

$$12) V_c = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} = \frac{5 \times 10^{-6} \times 16 + 10 \times 10^{-6} \times 13}{15 \times 10^{-6}} \quad (1)$$

$$= \frac{210}{15} \text{ V} = 14 \text{ volt} \quad (1)$$

Section C



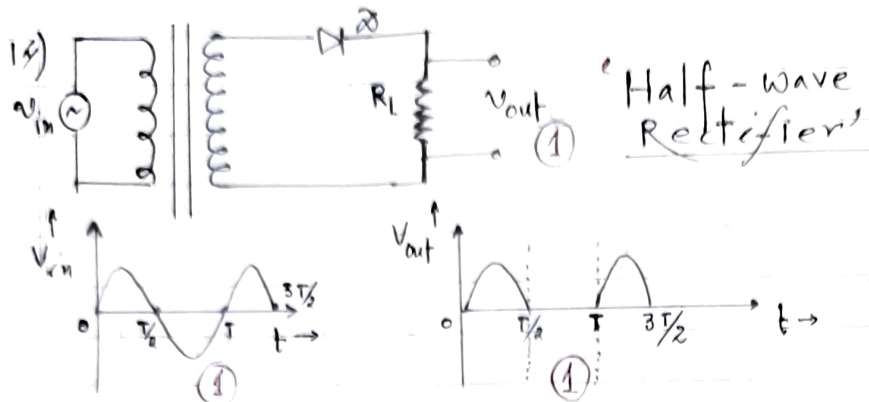
①) $T = qEl \sin \theta \approx qEl\theta$ (as θ is small)

$$I_{\text{net}} = I \propto \left[m \left(\frac{l}{2} \right)^2 + m \left(\frac{l}{2} \right)^2 \right] \propto$$

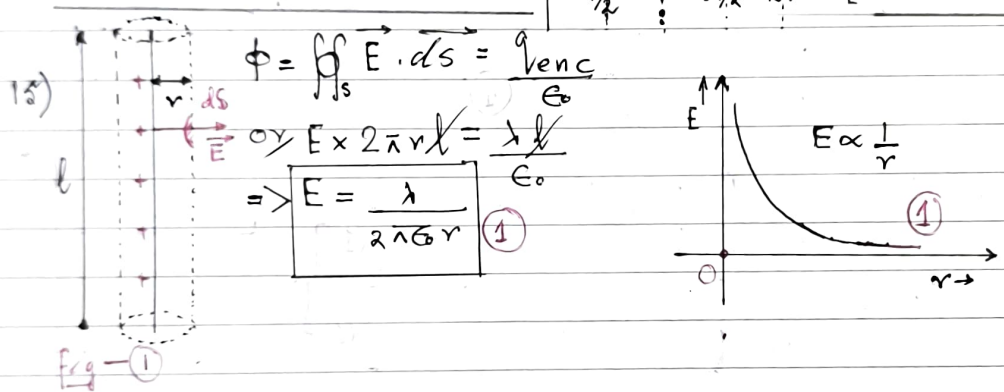
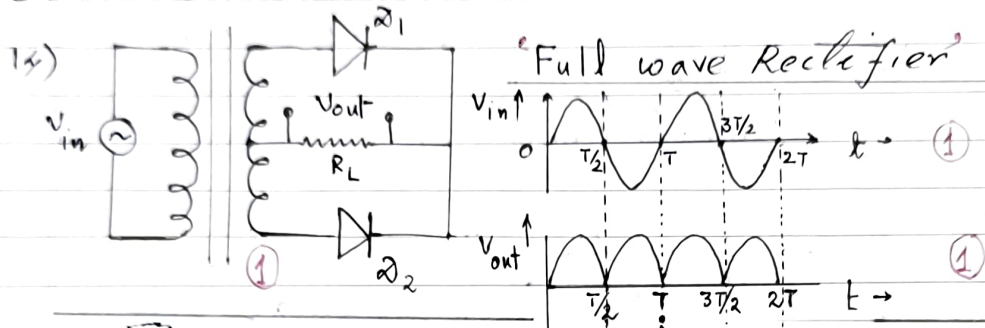
In Equilibrium,

$$\textcircled{1} \begin{cases} qE\ell = \frac{m\ell^2}{2} \frac{d\tilde{\theta}}{dt^2} \text{ or } \frac{d\tilde{\theta}}{dt^2} = \frac{2qE\ell}{m\ell} = \tilde{\omega}^2\theta \\ \therefore \omega = \sqrt{\frac{2qE}{m}} = \frac{2\pi}{T} \text{ or } T = \frac{2\pi}{\omega} \sqrt{\frac{m\ell}{2qE}} \end{cases}$$

$$\text{Reqd. time} = \frac{T}{4} = \frac{\pi}{2} \sqrt{\frac{m l}{2 q E}} \quad (1)$$



OR



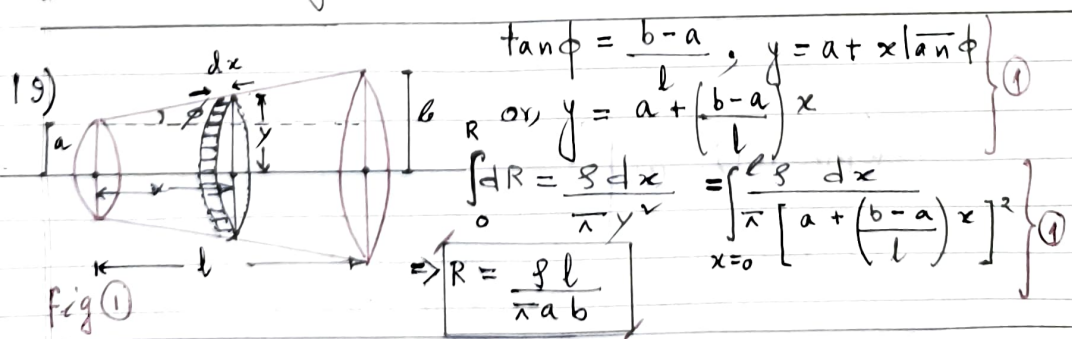
16) Fig ①, Proof ②

17) Fig ②, $\phi = 180^\circ$ or π radian ①

⑱ Proof ②, Significance: $I_{av} \times \frac{T}{2} = \int_0^{T/2} i_0 \sin \omega t dt = q$ ①

OR

⑱ Proof ②, Significance: $I_{rms}^2 R \frac{T}{2} = \int_0^{T/2} i_0^2 \sin^2 \omega t R dt = H$ ①



↳ Integration:

$$\text{Let } a + \frac{(b-a)}{l} x = \bar{x} \quad \text{or, } \frac{(b-a)}{l} dx = d\bar{x}$$

$$\text{When, } x=0, \bar{x}=a \quad \text{and } x=l, \bar{x}=b$$

$$\therefore R = \frac{\rho}{\pi} \frac{l}{(b-a)} \int_a^b \frac{d\bar{x}}{\bar{x}} = \frac{\rho l}{\pi (b-a)} \left[-\frac{1}{\bar{x}} \right]_a^b = \frac{\rho l}{\pi (b-a)} \left(\frac{1}{a} - \frac{1}{b} \right)$$

$$\text{or, } R = \frac{\rho l}{\pi (b-a)} \times \frac{(b-a)}{ab}$$

$$\text{or, } R = \frac{\rho l}{\pi ab}$$

$$20a) E = E_0 \cos(2\pi f t), \quad f = \frac{31.4}{2\pi} = 50 \text{ Hz}$$

$$X_L = 2\pi f L = 2\pi \times 50 \times 0.1 = 31.4 \Omega$$

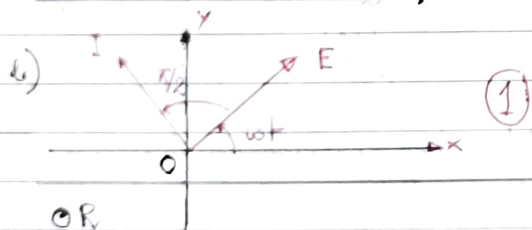
$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 25 \times 10^{-6}} = 127.4 \Omega$$

$$(i) \bar{z} = \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{25^2 + (127.4 - 31.4)^2} = 99.2 \Omega$$

$$(ii) \tan \phi = \frac{X_C - X_L}{R} = \frac{96}{25} = 3.84 \quad \text{or, } \phi = 75.4^\circ$$

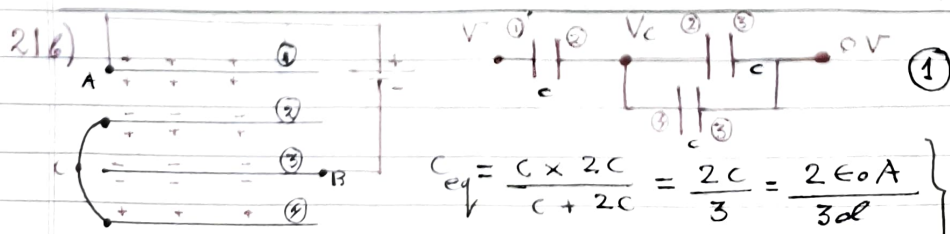
is the phase angle by which i leads E .

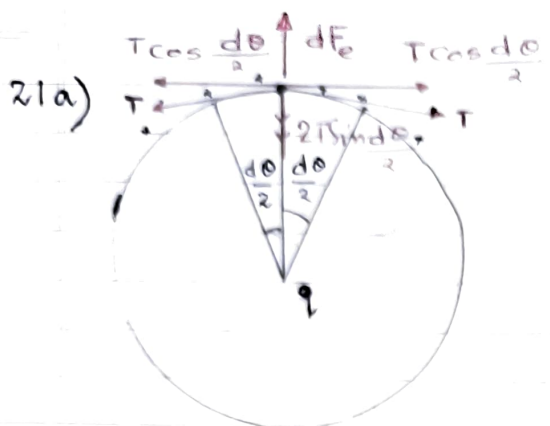
$$(iii) \text{Power factor } \cos \phi = \cos 75.4^\circ = 0.25.$$



$$20a) \text{ Fig (1), Proof (2). } B = \frac{\mu_0 i}{4\pi r} [\sin \phi_1 + \sin \phi_2]$$

$$b) M = \frac{\mu_0 N_1 N_2 A}{l} \quad \text{AT/steps (2)}$$





$$\lambda = \frac{Q}{2\pi R}, \quad dq = \lambda dl = \lambda R d\theta$$

$$\text{or, } dq = \frac{Q}{2\pi} \times d\theta$$

$$\Rightarrow dq = \frac{Q d\theta}{2\pi} \quad (1)$$

$$\left. \begin{aligned} dF_e &= \frac{1}{4\pi\epsilon_0} \frac{q \times Q d\theta}{R^2} \\ &= \frac{Q q d\theta}{8\pi^2 \epsilon_0 R^2} \end{aligned} \right\} (1)$$

In Equilibrium: $dF_e = 2T \sin \frac{d\theta}{2} = T d\theta$

$$\text{or, } \frac{Q q d\theta}{8\pi^2 \epsilon_0 R^2} = T d\theta$$

$$\text{or, } T = \frac{Q q}{8\pi^2 \epsilon_0 R^2} \quad (1)$$

OR

$$21a) \frac{-13.6}{n^2} = -3.4 \quad \text{or, } n=2 \quad (1)$$

By Bohr's Quantisation condition

$$\left. \begin{aligned} mvr &= \frac{n h}{2\pi} = 2 \times \frac{h}{2\pi} = \frac{h}{\pi} \\ \therefore L &= \frac{h}{\pi} \end{aligned} \right\} (1)$$

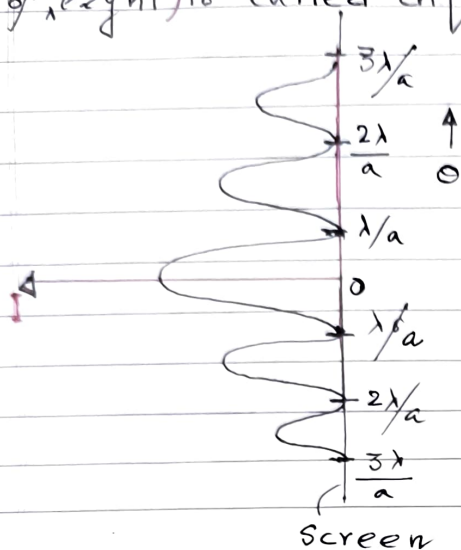
6) $\frac{dN}{dt} = -\lambda N + \alpha \quad (1)$ May be obtained by the given condition.

$$\text{or, } \int_{N_0}^N \frac{dN}{\alpha - \lambda N} = \int_0^t dt \quad \text{or, } \left[\frac{-1}{\lambda} \log_e (\alpha - \lambda N) \right]_{N_0}^N = t$$

$$\text{or, } \frac{1}{\lambda} \log_e \left| \frac{\alpha - \lambda N_0}{\alpha - \lambda N} \right| = t \quad (1)$$

$$\text{or, } N = \left\{ \alpha - (\alpha - \lambda N_0) e^{-\lambda t} \right\} \frac{1}{\lambda} \quad (1)$$

22a) The phenomenon of bending of light round the sharp corners and spreading into the region of geometrical shadow (when the size of the obstacle is comparable to the wavelength of ^{incident} light) is called diffraction of light. (1)



22b) $\mu = \tan \beta$ or, $\beta = \tan^{-1} \mu$ or, $\beta = 60^\circ$ (1)

OR

22a) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ or, $\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{f+u}{uf}$ or, $v = \frac{uf}{u+f}$ (1)

or, $\frac{dv}{dt} = \frac{(u+f)f \frac{du}{dt} - uf \cdot \frac{du}{dt}}{(u+f)^2}$

(1) or, $\frac{dv}{dt} = \left[\frac{f}{u+f} \right] \frac{du}{dt}$

as, for a convex lens f is +ve, u is (-)ve

$\Rightarrow \frac{dv}{dt} > \frac{du}{dt}$

$\therefore (f+u) < f \Rightarrow \frac{f^2}{(u+f)^2} > 1$ (1)

Thus the image of the ant moves with a greater speed than that of the ant.

22b) $\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$ (1)

or, $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \left[-\frac{1}{R} \right] \right)$

or, $\frac{1}{f} = (\mu - 1) \times \frac{2}{R}$, ATP, $R = f$

or, $\frac{1}{f} = \frac{2(\mu - 1)}{f}$ or, $\mu - 1 = \frac{1}{2}$ or, $\mu = 1.5$ (1)

